

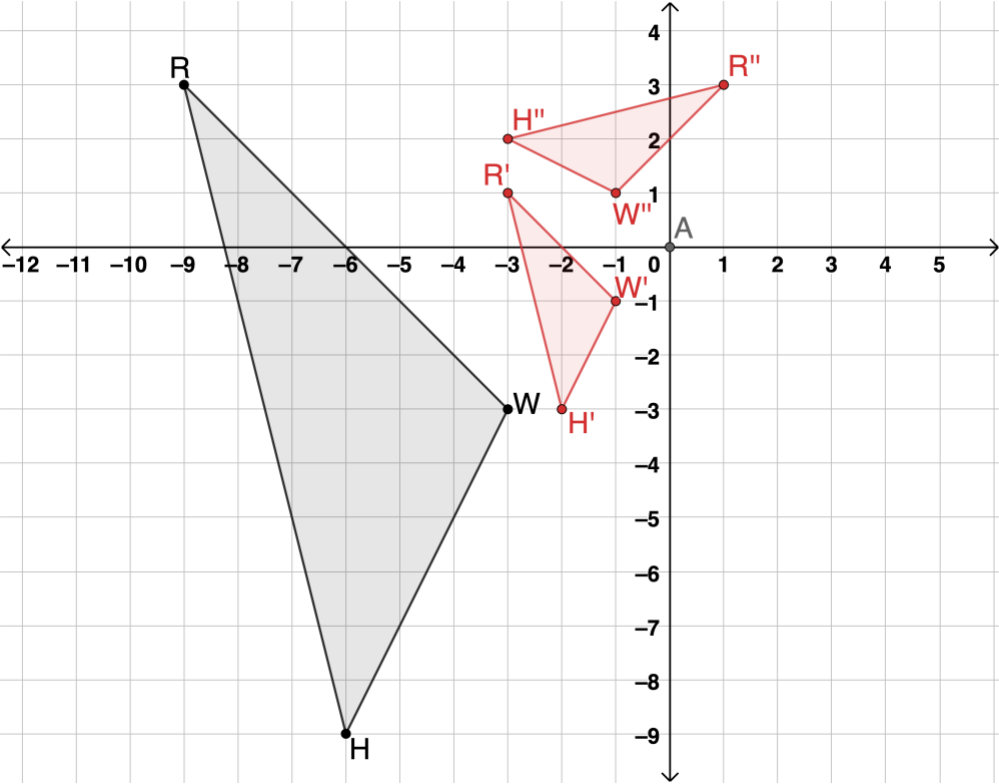
## Geometry Honors Unit 6 Part 1 Assessment Guide

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| <p style="text-align: center;"><b>General Information</b></p>      | <p>This assessment covers the benchmarks from lessons 1-6. In Unit 1 students explored dilations and in Unit 4 students focused on the rigid motions found in MA.912.GR.2.1, MA.912.GR.2.3, and MA.912.GR.2.5. Unit 6, lessons 1-6 allow the students to spiral their learning while expanding these benchmarks to include dilations. Teachers can use this assessment after lessons 6 or use the questions from this assessment and part 2 to build one assessment for entire unit.</p> <p>The benchmarks in this unit have similarities. For some of the questions on the assessment the question is asking the student to show knowledge on content that is shared between more than one benchmark, as noted. The nuances of the benchmarks can be found in the language of each benchmark. For the student the application of their knowledge may not seem different.</p> <p>The assessment was created so that teachers have information on student weaknesses on the various transformations while also assessing the similarity transformation of dilation. As the content of rigid motions has been spiraled within the unit, the unit assessment can provide teachers with an understanding of students have grasped both types of transformations. Teachers are encouraged to monitor student understanding of the single transformations as well as when there is an incorporation of a sequence of transformations. If students can understand a single transformation but struggle with a sequence of transformations, then the teacher can offer various strategies focused on understanding how to complete the sequence. Such strategies may include applying each transformation to one vertex at a time or cutting out and physically manipulating the figure, marking the vertices after each transformation.</p> <p>The honors benchmarks MA.912.GR.6.5 and MA.912.LT.4.8 are paired together in question #. Students first complete a transformation to map the two given circles onto each other and then show that they are similar.</p> |
| <p style="text-align: center;"><b>Calculator Information</b></p>   | <p>All questions are calculator neutral. Students may be given the use of a calculator and it will not have influence on their ability to show their knowledge of the content.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p style="text-align: center;"><b>Total Recommended Points</b></p> | <p>The total recommended points in this guide are 100 points.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Question         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Benchmark(s)                                                     | MA.912.GR.2.1                          |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------|-----------------------|------------------|---------------------------|------------------------------------|------------------|-----------------------------------------|----------------------------|------------------|----------------------------------------|----------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>Complete the statement.</p> <p>Triangle <div><div><div><input type="radio"/> <i>DNS</i></div><div><input type="radio"/> <i>KRB</i></div></div></div> is the result of dilating triangle <div><div><div><input type="radio"/> <i>DNS</i></div><div><input type="radio"/> <i>KRB</i></div></div></div> using a scale factor of 3 centered at <div><div><div><input type="radio"/> <i>H.</i></div><div><input type="radio"/> <i>L.</i></div><div><input type="radio"/> <i>W.</i></div></div></div></p> | <p><b>Recommended Scoring</b></p> <p><b>(4 Total Points)</b></p> | <p>No partial credit is suggested.</p> |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
| Question         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Benchmark(s)                                                     | MA.912.GR.2.1<br>MA.912.GR.2.3         |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
|                  | <table><tr><td></td><th>Written Description</th><th>Algebraic Description</th></tr><tr><th>Transformation 1</th><td>Translation left 14, up 7</td><td><math>(x,y) \rightarrow (x - 14,y + 7)</math></td></tr><tr><th>Transformation 2</th><td>Rotation 90° clockwise about the origin</td><td><math>(x,y) \rightarrow (y,-x)</math></td></tr><tr><th>Transformation 3</th><td>Dilation of 2.5 centered at the origin</td><td><math>(x,y) \rightarrow (2.5x, 2.5y)</math></td></tr></table>             |                                                                  | Written Description                    | Algebraic Description | Transformation 1 | Translation left 14, up 7 | $(x,y) \rightarrow (x - 14,y + 7)$ | Transformation 2 | Rotation 90° clockwise about the origin | $(x,y) \rightarrow (y,-x)$ | Transformation 3 | Dilation of 2.5 centered at the origin | $(x,y) \rightarrow (2.5x, 2.5y)$ | <p><b>Recommended Scoring</b></p> <p><b>(15 Total Points)</b></p> | <p>Consider 5 points for each transformation, distributed evenly to the written description and the algebraic description. Consider partial credit if a student’s written and algebraic match but it contains a minor error related to the transformation.</p> |
|                  | Written Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Algebraic Description                                            |                                        |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
| Transformation 1 | Translation left 14, up 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $(x,y) \rightarrow (x - 14,y + 7)$                               |                                        |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
| Transformation 2 | Rotation 90° clockwise about the origin                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $(x,y) \rightarrow (y,-x)$                                       |                                        |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |
| Transformation 3 | Dilation of 2.5 centered at the origin                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $(x,y) \rightarrow (2.5x, 2.5y)$                                 |                                        |                       |                  |                           |                                    |                  |                                         |                            |                  |                                        |                                  |                                                                   |                                                                                                                                                                                                                                                                |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                          | Benchmark(s)                                                                                                                                                         | MA.912.GR.2.1<br>MA.912.GR.2.3 |                          |                          |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
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| <table><tr><th>Transformation of quadrilateral <i>RDWY</i></th><th><i>LDPC</i></th><th><i>TDMH</i></th><th><i>BXNK</i></th><th><i>SVGP</i></th><th>none</th></tr><tr><td>translation down 2</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>translation right 6</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>translation right 3 down 4</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>dilation by a factor of 2 centered at point <i>D</i></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>dilation by a factor of <math>\frac{1}{2}</math> centered at the origin</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>reflection across <i>y</i> –axis</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>reflection across <math>x = -3</math></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>rotation 180° about point <i>D</i></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr><tr><td>rotation 180° about the origin</td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr></table> | Transformation of quadrilateral <i>RDWY</i>                | <i>LDPC</i>                                                                                                                                                          | <i>TDMH</i>                    | <i>BXNK</i>              | <i>SVGP</i>              | none | translation down 2 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | translation right 6 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | translation right 3 down 4 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | dilation by a factor of 2 centered at point <i>D</i> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | dilation by a factor of $\frac{1}{2}$ centered at the origin | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | reflection across <i>y</i> –axis | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | reflection across $x = -3$ | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | rotation 180° about point <i>D</i> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | rotation 180° about the origin | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <div>Recommended Scoring</div> <div>(27 Total Points)</div> | Consider 3 points for each correct match. |
| Transformation of quadrilateral <i>RDWY</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>LDPC</i>                                                | <i>TDMH</i>                                                                                                                                                          | <i>BXNK</i>                    | <i>SVGP</i>              | none                     |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| translation down 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| translation right 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| translation right 3 down 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| dilation by a factor of 2 centered at point <i>D</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| dilation by a factor of $\frac{1}{2}$ centered at the origin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| reflection across <i>y</i> –axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| reflection across $x = -3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| rotation 180° about point <i>D</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| rotation 180° about the origin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <input type="checkbox"/>                                   | <input type="checkbox"/>                                                                                                                                             | <input type="checkbox"/>       | <input type="checkbox"/> | <input type="checkbox"/> |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4a                                                         | Benchmark(s)                                                                                                                                                         | MA.912.GR.2.3                  |                          |                          |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |
| <div><input type="checkbox"/> <math>\triangle BRT</math> maps to <math>\triangle YNH</math> using <math>(x,y) \rightarrow (-y,x)</math> then <math>(x,y) \rightarrow (x-3,y-4)</math> then <math>(x,y) \rightarrow (2x,2y)</math></div> <div><input type="checkbox"/> <math>\triangle BRT</math> maps to <math>\triangle YNH</math> using <math>(x,y) \rightarrow (x-7,y-3)</math> then <math>(x,y) \rightarrow (x,-y)</math> then <math>(x,y) \rightarrow (2.5x,2.5y)</math></div> <div><input type="checkbox"/> <math>\triangle BRT</math> maps to <math>\triangle YNH</math> using <math>(x,y) \rightarrow (-y,-x)</math> then <math>(x,y) \rightarrow (5x,5y)</math> then <math>(x,y) \rightarrow (x,-y)</math></div> <div><input type="checkbox"/> <math>\triangle YNH</math> maps to <math>\triangle BRT</math> using <math>(x,y) \rightarrow (x+6,y)</math> then <math>(x,y) \rightarrow (x,-y)</math> then <math>(x,y) \rightarrow (0.5x,0.5y)</math></div> <div><input type="checkbox"/> <math>\triangle YNH</math> maps to <math>\triangle BRT</math> using <math>(x,y) \rightarrow (0.4x,0.4y)</math> then <math>(x,y) \rightarrow (x,-y)</math> then <math>(x,y) \rightarrow x+7,y+3)</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>Recommended Scoring</div> <div>(8 Total Points)</div> | Consider 4 points for each correct selection. Consider deducting 3 points for each additional selection. Consider no points if a student selects all of the choices. |                                |                          |                          |      |                    |                          |                          |                          |                          |                          |                     |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                                      |                          |                          |                          |                          |                          |                                                              |                          |                          |                          |                          |                          |                                  |                          |                          |                          |                          |                          |                            |                          |                          |                          |                          |                          |                                    |                          |                          |                          |                          |                          |                                |                          |                          |                          |                          |                          |                                                             |                                           |

| Question                                                                                                                                                                                                                                                                                                                  | 4b | Benchmark(s)                                          | MA.912.GR.2.8                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p>The mapping of triangle <math>BRT</math> to triangle <math>YNH</math> includes a dilation, so therefore the triangles are similar.</p> <p style="text-align: center;">OR</p> <p>The mapping of triangle <math>YNH</math> to triangle <math>BRT</math> includes a dilation, so therefore the triangles are similar.</p> |    | <b>Recommended Scoring</b><br><b>(5 Total Points)</b> | Consider 5 points for a correct justification that includes an understanding of dilation. |
| Question                                                                                                                                                                                                                                                                                                                  | 5  | Benchmark(s)                                          | MA.912.GR.2.1<br>MA.912.GR.2.3                                                            |
| <p>Reflect <math>\triangle RHB</math> across the <math>x</math>-axis, then dilate using a scale of 4.5 centered at the origin.</p>                                                                                                                                                                                        |    | <b>Recommended Scoring</b><br><b>(8 Total Points)</b> | Consider giving students 4 points for each correct transformation in the sequence.        |

| Question                                                                            | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Benchmark(s)                                                     | MA.912.GR.2.5                                                                                                                                                                                                                                                          |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
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|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b>Recommended Scoring</b></p> <p><b>(8 Total Points)</b></p> | <p>Consider giving students 4 points for the correct dilation and 4 points for the correct rotation. If students do not show the dilation but do have the correct result, then the student should receive full credit. Not all students need to show the mid-step.</p> |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
| Question                                                                            | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Benchmark(s)                                                     | MA.912.GR.2.2                                                                                                                                                                                                                                                          |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
|                                                                                     | <table border="1" data-bbox="149 1060 1234 1448"> <thead> <tr> <th>Sequence of Transformations</th><th>Preserves Distance</th><th>Preserves Angle Measure</th><th><math>H''R''N''C''</math> is similar but not congruent to <math>HRNC</math></th></tr> </thead> <tbody> <tr> <td><math>(x, y) \rightarrow (-y, x)</math> then <math>(x, y) \rightarrow (3.5x, 3.5y)</math></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr> <tr> <td><math>(x, y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)</math> then <math>(x, y) \rightarrow (x, -y)</math></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr> <tr> <td><math>(x, y) \rightarrow (x - 1, y + 5)</math> then <math>(x, y) \rightarrow (x, y)</math></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr> <tr> <td><math>(x, y) \rightarrow (-x, -y)</math> then <math>(x, y) \rightarrow (x + 6, y - 2)</math></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td><td><input type="checkbox"/></td></tr> </tbody> </table> | Sequence of Transformations                                      | Preserves Distance                                                                                                                                                                                                                                                     | Preserves Angle Measure | $H''R''N''C''$ is similar but not congruent to $HRNC$ | $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (3.5x, 3.5y)$ | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | $(x, y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$ then $(x, y) \rightarrow (x, -y)$ | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | $(x, y) \rightarrow (x - 1, y + 5)$ then $(x, y) \rightarrow (x, y)$ | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | $(x, y) \rightarrow (-x, -y)$ then $(x, y) \rightarrow (x + 6, y - 2)$ | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <p><b>Recommended Scoring</b></p> <p><b>(16 Total Points)</b></p> | <p>Consider 4 points for the correct match for each sequence. Partial credit is not suggested.</p> |
| Sequence of Transformations                                                         | Preserves Distance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Preserves Angle Measure                                          | $H''R''N''C''$ is similar but not congruent to $HRNC$                                                                                                                                                                                                                  |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
| $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (3.5x, 3.5y)$                 | <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/>                                         | <input type="checkbox"/>                                                                                                                                                                                                                                               |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
| $(x, y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$ then $(x, y) \rightarrow (x, -y)$ | <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/>                                         | <input type="checkbox"/>                                                                                                                                                                                                                                               |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
| $(x, y) \rightarrow (x - 1, y + 5)$ then $(x, y) \rightarrow (x, y)$                | <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/>                                         | <input type="checkbox"/>                                                                                                                                                                                                                                               |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |
| $(x, y) \rightarrow (-x, -y)$ then $(x, y) \rightarrow (x + 6, y - 2)$              | <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/>                                         | <input type="checkbox"/>                                                                                                                                                                                                                                               |                         |                                                       |                                                                     |                          |                          |                          |                                                                                     |                          |                          |                          |                                                                      |                          |                          |                          |                                                                        |                          |                          |                          |                                                                   |                                                                                                    |

| Question | 8a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Benchmark(s)                                                     | MA.912.GR.6.5                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------|
|          | <p>Lakesha first dilates circle <math>H</math> with a center of dilation at point <math>H</math> using a scale factor of</p> <div data-bbox="300 232 424 581"> <ul style="list-style-type: none"> <li><input type="radio"/> 1.</li> <li><input type="radio"/> 2.</li> <li><input type="radio"/> 2.5.</li> <li><input type="radio"/> 3.</li> <li><input checked="" type="radio"/> 5.</li> <li><input type="radio"/> 5.5.</li> <li><input type="radio"/> 3.</li> <li><input type="radio"/> 5.</li> <li><input type="radio"/> 2.</li> <li><input type="radio"/> 2.</li> </ul> </div> <p>She then translates the circle</p> <div data-bbox="823 311 1190 511"> <ul style="list-style-type: none"> <li><input type="radio"/> right 11.</li> <li><input type="radio"/> right 6 down 3.</li> <li><input checked="" type="radio"/> right 11 down 4.</li> <li><input type="radio"/> right 8.5 down 2.5.</li> </ul> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b>Recommended Scoring</b></p> <p><b>(3 Total Points)</b></p> | <p>No partial credit is suggested.</p> |
| Question | 8b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Benchmark(s)                                                     | MA.912.LT.4.8<br>MA.912.GR.6.5         |
|          | <p>Assume that circle <math>H</math></p> <div data-bbox="443 695 606 792"> <ul style="list-style-type: none"> <li><input type="radio"/> is</li> <li><input checked="" type="radio"/> is not</li> </ul> </div> <p>similar to circle <math>M</math>. By definition of similarity,</p> <p>two figures are similar if and only if one figure can be obtained from the other by</p> <p>a</p> <div data-bbox="189 873 428 1053"> <ul style="list-style-type: none"> <li><input checked="" type="radio"/> dilation,</li> <li><input type="radio"/> rotation,</li> <li><input type="radio"/> reflection,</li> <li><input type="radio"/> translation,</li> </ul> </div> <p>or a sequence of transformations that includes a</p> <div data-bbox="161 1075 401 1255"> <ul style="list-style-type: none"> <li><input checked="" type="radio"/> dilation.</li> <li><input type="radio"/> rotation.</li> <li><input type="radio"/> reflection.</li> <li><input type="radio"/> translation.</li> </ul> </div> <p>Because one of the transformations is a</p> <div data-bbox="951 1075 1190 1255"> <ul style="list-style-type: none"> <li><input checked="" type="radio"/> dilation,</li> <li><input type="radio"/> rotation,</li> <li><input type="radio"/> reflection,</li> <li><input type="radio"/> translation,</li> </ul> </div> <p>the circles are mapped using</p> <div data-bbox="552 1276 978 1365"> <ul style="list-style-type: none"> <li><input type="radio"/> rigid motions.</li> <li><input checked="" type="radio"/> similarity transformations.</li> </ul> </div> <p>Therefore, circle <math>H</math></p> <p>is similar to circle <math>M</math>, which contradicts the given statement.</p> | <p><b>Recommended Scoring</b></p> <p><b>(3 Total Points)</b></p> | <p>No partial credit is suggested.</p> |

| Question                                                                                                                                                                                                                                                                            | 8c | Benchmark(s)                                                     | MA.912.GR.6.5                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------|----------------------------------------|
| <p>All circles <div><div><div><input type="radio"/> can</div><div><input type="radio"/> cannot</div></div></div> be mapped to each other using <div><div><div><input type="radio"/> rigid motions.</div><div><input type="radio"/> similarity transformations</div></div></div></p> |    | <p><b>Recommended Scoring</b></p> <p><b>(3 Total Points)</b></p> | <p>No partial credit is suggested.</p> |